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Pre-Generation Causal Drift Metric

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By

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Pre-Generation Causal Drift Metric for Hallucination Detection in Retrieval-Augmented Language Models

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Abstract

This study investigates whether hidden-states within transformer layers provide predictive signals of hallucination before it is generated in retrieval-augmented generation (RAG) systems. Five representation-space drift signals are combined into a unified detection framework: cosine drift, Mahalanobis distance, logit lens divergence, PCA deviation, and causal indirect effect (CIE). Experiments are conducted on two benchmark datasets, RAGTruth and HaluEval, spanning diverse hallucination scenarios. The composite detector achieves an AUROC of 0.6605 (95% CI: 0.6396–0.6814) on the test set, improving the attention entropy baseline by 1.08 AUROC points and closing 4.90% of the remaining gap toward the Lumina benchmark. Causal patching experiments across 147 bidirectional activation interventions show that late feed-forward network layers have statistically significant causal indirect effects in hallucination generation. Temporal analysis also shows that representation drift peaks at position $t - 2$, suggesting that hidden-state divergence occurs at least two tokens before the hallucinated output appears, which is further confirmed through Mann–Whitney significance tests.

Keywords: hallucination detection, retrieval-augmented generation, mechanistic interpretability, representation drift, causal patching, RAGTruth

1 Introduction

1.1 Motivation

Although Retrieval-Augmented Generation (RAG) reduces factual errors by fact-checking information from external documents, hallucinations can still occur. This is because sometimes the model misunderstands the retrieved text, ignores parts of it, or depends more on its pre-trained knowledge [1]. Most existing hallucination detectors focus only on the final output using methods such as confidence scores or semantic mismatch signals to detect errors. This, however, cannot explain how or where hallucinations develop inside the model which limits interpretability [2]. A more detailed mechanistic approach is then needed to follow how the model’s internal representations change across

layers and identify the semantic shifts that happen before a hallucinated response is generated [3].

1.2 Research Question

“Do hidden-state dynamics within transformer layers provide predictive signals of hallucination before it manifests in generated text?”

Existing research identifies important transformer components and internal representations that cause hallucinations but does not directly link layer-level semantic drift to hallucination prediction in retrieval-augmented language models.

1.3 Contributions

- **Five-signal composite:** We combine cosine drift, Mahalanobis distance, logit lens divergence, PCA deviation, and causal indirect effect into a unified hidden-state hallucination detec-

tion framework.

- **Layer localisation:** Layer-wise profiling identifies hallucination-sensitive regions, with strongest separability concentrated in mid-layers (10-14), showing where internal semantic drift is most visible.
- **Causal patching:** We perform bidirectional activation patching with 147 experiments per direction, showing causal indirect effects and confirming late FFN layers are the reason why hallucination occurs.
- **Temporal precedence:** Representation drift peaks early in $t - 2$, with Mann-Whitney tests confirming statistically significant pre-generation divergence. This shows that hallucination emerges before output token generation.
- **SOTA gap:** Our composite improves over the attention-entropy baseline by 1.08 points, closing 4.90% of the remaining performance gap toward the current Lumina benchmark.

1.4 Paper Organisation

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 formalises all five metrics and CIE. Section 4 describes the full pipeline. Section 5 covers datasets. Section 6 details the experimental setup. Sections 7–11 report all experiments. Section 12 analyses findings and limitations. Section 13 concludes.

2 Related Work

2.1 Hallucination in Large Language Models

Hallucinations in large language models (LLMs) are generated texts that are either unsupported, factually incorrect, or inconsistent with the provided context. Recent work has identified two distinct types of hallucinations and analyzed their differences. Intrinsic hallucinations are largely centered on responses that contradict the source evidence, whereas extrinsic hallucinations introduce unverifiable information beyond any provided source. Benchmarks such as RagTruth and HaluEval provide large-scale evaluation datasets to analyze hallucination behavior in both standalone and retrieval-augmented settings [2,3]. SelfCheck-GPT further helps show that consistency-based un-

certainty signals can detect hallucinated generations without external supervision [4]. However, these approaches remain output-centric and cannot explain the internal mechanisms that generate hallucinations.

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) improves language models by adding a retrieval component that uses external documents. This component helps the model generate responses better based on factual information [1]. REALM expanded this idea by introducing retrieval-aware pre-training, where the model learns to use retrieved documents during training itself [5]. While Atlas showed that retrieval-based generation can still perform well even with limited labeled data [6]. Despite this, hallucinations still occur in RAG systems. This is not only because retrieval may fail to return relevant documents, but also because the retrieved information can conflict with the knowledge already stored in the model from pre-training. These conflicts are not always visible in the final output and may instead appear as semantic drift in the model’s internal hidden-state representations across transformer layers. This makes it important to study the model’s internal layer behavior, rather than only the final output probabilities.

2.3 Mechanistic Interpretability

Mechanistic interpretability aims to understand how transformers make decisions by studying their internal representations. Studies on transformer circuits show that model behavior comes from interactions between attention heads and feed-forward layers working together during reasoning [7]. The logit lens examines hidden states across layers by projecting them into vocabulary space. This shows how predictions become more refined through the network. Other methods such as causal tracing and ROME identify factual knowledge in specific layers and show that edits can change stored facts without affecting the overall behavior of the model [8]. Geva et al. also found that FFN layers are memory systems that store and retrieve factual texts [9]. Similarly, activation steering method shows that that hidden states contain meaningful semantic information because changing hidden-state activations can influence model behaviour [10]. However, attention weights alone

are not always reliable explanations for model decisions [11]. Together, these studies help explain how factual knowledge is represented inside transformers, but existing work still does not directly connect hidden-state drift across layers to hallucination prediction in retrieval-augmented systems.

2.4 Hallucination Detection in RAG

Recent hallucination detection methods for RAG systems combine uncertainty estimation with analysis of the model’s internal representations. These methods, which are primarily based on attention entropy and confidence scores, provide simple output-level signals for detecting hallucinations. However, they remain limited because they do not explain how the model internally processes retrieved information. One study, ReDeEP approaches this problem using mechanistic interpretability. It shows that hallucinations occur when Knowledge FFNs in later transformer layers overemphasise parametric knowledge, meanwhile the copying heads fail to preserve retrieved context. The study achieved around 0.82 AUROC, concluding that later FFN layers are strongly linked to hallucinations in RAG models [12]. Moreover, attribution-based reasoning frameworks analyse interaction between internal representations and retrieved evidence during generation [13]. LUMINA builds on this idea by measuring the model’s reliance on external context across layers and tracking how token predictions change through the network. LUMINA achieved around 0.87 AUROC on HalluRAG showing that analysing layer-wise retrieved context can provide stronger and more interpretable hallucination detection signals than confidence-based approaches alone. FaithJudge also highlights the importance of strong evaluation methods and human annotations for measuring faithfulness [14]. Together, these studies demonstrate a shift toward mechanistic and attribution-based approaches that better explain how hallucinations emerge in RAG systems.

Gap Summary

Existing methods can detect hallucinations or identify important transformer components, but they still do not explain when semantic drift begins during generation or how it leads to hallucinated outputs. ReDeEP and LUMINA both separate contextual and parametric knowledge signals, but

they treat them as fixed scores instead of studying how representations change across layers over time. They also do not track hidden-state divergence throughout generation or establish whether semantic drift happens before hallucinations appear in the output. This paper addresses that gap by combining representation-space drift analysis, layer localization, temporal analysis, and causal intervention. The problem is non-trivial because hallucination signals are distributed across transformer layers and depend on interactions between attention heads and FFN modules throughout the forward pass.

3 Problem Formulation

3.1 Notation and Setup

Let $\mathcal{M}$ be the language model with L transformer layers, q the query, $\mathcal{D} = \{d_1, \dots, d_k\}$ retrieved documents, $y = (y_1, \dots, y_T)$ the response, and $h_t^{(\ell)}(\cdot) \in \mathbb{R}^d$ the hidden state at layer ℓ and token t conditioned on the given context:

$$P_{\mathcal{M}}(y | q, \mathcal{D}) = \prod_{t=1}^T P_{\mathcal{M}}(y_t | y_{<t}, q, \mathcal{D}). \quad (1)$$

Definition 3.1 (Hallucination). Token y_t is hallucinated if it is not entailed by any $d_i \in \mathcal{D}$ and is factually incorrect with respect to ground-truth knowledge $\mathcal{K}$.

3.2 Signal 1 — Cosine Drift

Definition 3.2 ($\delta_t^{(\ell)}$).

$$\delta_t^{(\ell)} = 1 - \frac{h_t^{(\ell)}(\mathcal{D}) \cdot h_t^{(\ell)}(\emptyset)}{\|h_t^{(\ell)}(\mathcal{D})\| \|h_t^{(\ell)}(\emptyset)\|}. \quad (2)$$

High cosine drift at a given layer indicates consecutive hidden states are diverging, suggesting the model’s internal representation is drifting from context-grounded processing toward parametric hallucination.

3.3 Signal 2 — Mahalanobis Distance

Definition 3.3 ($m_t^{(\ell)}$).

$$m_t^{(\ell)} = \sqrt{(h_t^{(\ell)} - \mu_\ell)^\top \Sigma_\ell^{-1} (h_t^{(\ell)} - \mu_\ell)}, \quad (3)$$

where μ_ℓ and Σ_ℓ are estimated on the **training split only** (no validation or test hidden states).

A high Mahalanobis score indicates the hidden state lies outside the covariance structure of faithful generation, signaling distributional deviation. All μ_ℓ and Σ_ℓ estimates use training-split tokens only to prevent data leakage.

3.4 Signal 3 — Logit Lens Divergence

Definition 3.4 ($\Lambda_t^{(\ell)}$). Project hidden state through the unembedding matrix $W_U \in \mathbb{R}^{|V| \times d}$:

$$\hat{P}_t^{(\ell)}(\cdot) = \text{softmax}(W_U h_t^{(\ell)}(\cdot)). \quad (4)$$

$$\Lambda_t^{(\ell)} = D_{\text{KL}}(\hat{P}_t^{(\ell)}(\mathcal{D}) \parallel \hat{P}_t^{(\ell)}(\emptyset)). \quad (5)$$

This KL divergence captures how token-level predictions in intermediate layers shift toward the final output distribution as retrieved context is progressively integrated.

3.5 Signal 4 — PCA Deviation

Definition 3.5 ($\rho_t^{(\ell)}$). Let $V_\ell \in \mathbb{R}^{d \times r}$ be the top- r principal components of **training-split** faithful hidden states at layer ℓ :

$$\rho_t^{(\ell)} = \|h_t^{(\ell)}(\mathcal{D}) - V_\ell V_\ell^\top h_t^{(\ell)}(\mathcal{D})\|_2. \quad (6)$$

High deviation indicates the hidden state lies outside the subspace characteristic of faithful generation. V_ℓ is estimated on training split only.

3.6 Signal 5 — Causal Indirect Effect

Definition 3.6 (CIE_c). For component c (attention head or FFN sublayer), patch its activations from a clean faithful run into a corrupted hallucinated run:

$$\text{CIE}_c = P_{\mathcal{M}}^{\text{patch}(c)}(y_t^*) - P_{\mathcal{M}}^{\text{corrupt}}(y_t^*), \quad (7)$$

where y_t^* is the ground-truth faithful token.

Patching is performed bidirectionally—both faithful to hallucinated and hallucinated to faithful directions. A minimum of 50 experiments per component type establishes statistical reliability, following the causal intervention framework of Meng et al. [8].

3.7 Composite Metric

The composite is built **incrementally** — each signal evaluated independently first, then added one at a time (required for E1 and E2). Layer signals are aggregated over top-3 most informative layers

$\mathcal{L}^*$ (identified on the validation set from the layer-profile plot):

$$s_t = \sum_{i=1}^5 w_i \tilde{\phi}_i(t) + w_6 \widetilde{\text{CIE}}_t, \quad (8)$$

where $\tilde{\phi}_i \in [0, 1]$ are min-max normalised signals, $\sum_i w_i = 1$, weights chosen on the validation set. Decision rule: $\hat{y}_t = \mathbf{1}[s_t > \tau]$.

3.8 Theoretical Justification

Hypothesis 3.1. The Track B composite achieves AUROC ≥ 0.70 on the RAGTruth held-out test set (bootstrap 95 % CI, $N = 1000$), outperforms both Attention entropy and Logit confidence baselines, at least one drift metric peaks at position $t-2$ or earlier relative to the first hallucinated token ($p < 0.05$, Mann-Whitney U), and CIE is significant in ≥ 2 component types ($p < 0.05$) across ≥ 50 patching experiments.

Geva et al. [9] demonstrates that FFN sublayers function as key-value memories storing factual associations in mid-to-late transformer layers. Meng et al. [8] further shows that such parametric knowledge can be localized and edited, highlighting its vulnerability under conflicting retrieval signals in RAG settings. When retrieved context conflicts with stored associations, hidden states drift from the faithful distribution prior to generation. Mahalanobis distance and cosine drift capture this deviation in representation space, while logit lens projects intermediate states into vocabulary space for interpretability, enabling pre-generation detection of hallucination.

4 Methodology

4.1 System Architecture

The system uses a RAGtruth provided context with GPT-2 medium to extract layer-wise hidden states, attention, FFN outputs, and logits. These extracted representations are used to calculate hallucination signals like cosine drift, Mahalanobis distance, logit lens divergence, PCA deviation, and CIE. Normalized metric scores are combined into a composite hallucination score. If the composite score crosses the threshold, the response is classified as hallucinated; otherwise, it is classified as faithful.

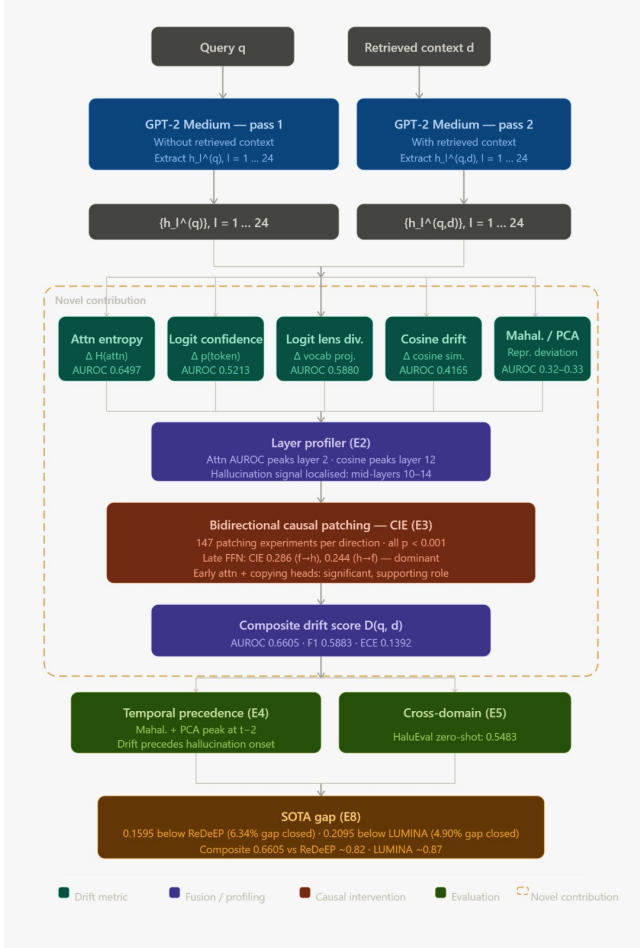


Figure 1: Track B pipeline. Two forward passes extract hidden states at all layers. Five drift signals are computed. A layer-profile identifies top-3 layers. CIE from bidirectional causal patching is fused into the composite. Dashed box = novel contribution.

4.2 Step 1 — Retrieval

This work uses a retrieval augmented setup based on the provided RAGTruth context from `source_info.jsonl` joined by `source_id.matched`. Therefore, the Retrieval is top $k = 1$. The context window is created by concatenating the given question, context and generated response into a single input order for GPT-2-medium hidden-state extraction

4.3 Step 2 — Hidden-State Extraction

Register_forward_hook is applied on all transformer layers during the forward pass to obtain layer-wise token representations. For each token t and layer l , we collected hidden states under both context-conditioned and no-context settings, denoted by $h_t^{(\ell)}(\mathcal{D})$ and $h_t^{(\ell)}(\emptyset)$ respectively. The ref-

erence statistics Σ_ℓ , V_ℓ , and μ_ℓ were calculated using the training split. For memory management, tensors were separated, shifted to CPU, processed in small batches, and the required statistics for scoring were retained.

4.4 Step 3 — Drift Metric Computation

For every single token t and layer l , we calculate five drift signals: Mahalanobis distance, cosine drift, PCA deviation, attention entropy and logit-lens divergence. Scores were estimated across all transformer layers and aggregated over the top-3 most associated layers $\mathcal{L}^*$. Token-level drift scores were totaled using the mean across these layers.

4.5 Step 4 — Layer Profile Analysis

Layer profiling is calculated by point-biserial correlation between cosine drift and binary hallucination label. Point biserial is used to measure between a continuous score such as cosine drift and binary label, such as hallucinated vs faithful. In each layer the coefficient r quantified how strong the drift is associated with hallucination. Layers are ranked by $|r|$ and top-3 layers with the highest correlation are $\mathcal{L}^* = \{11, 12, 10\}$

4.6 Step 5 — Causal Patching Protocol

147 example pairs were made by grouping samples with the same source_id from the training set. For each source, a faithful activation was patched into hallucinated examples and hallucinated activations were patched into faithful examples. Interventions were evaluated in both direction. This two-way approach checks if the chosen parts can reduce or increase hallucination-like behaviour. Four Component groups are analysed: early attention heads (layers 1-25%), mid FFN layers (26-75%), late FFN layers (76-100%), and copying heads (last 25%). The wilcoxon test was used to check if the two patching directions produced different causal effects. A lower P value indicates it has directional sensitive causal influence on hallucination behaviour as the two patching direction gives different CIE scores.

4.7 Step 6 — Composite Fusion

The weights $\{w_i\}$ are chosen corresponding to validation AUROC, so stronger metrics contribute more and the threshold τ is chosen to maximize F1 on validation data.

Algorithm 1 Track B Composite Score Computation**Require:** q , $\mathcal{D}$, $\mathcal{M}$, top-3 layers $\mathcal{L}^*$, weights $\{w_i\}$, τ **Ensure:** Per-token labels $(\hat{y}_1, \dots, \hat{y}_T)$

```

1: Extract  $h_t^{(\ell)}(\mathcal{D})$  and  $h_t^{(\ell)}(\emptyset)$  for all  $t, \ell$ 
2: for  $t = 1$  to  $T$  do
3:   for  $\ell \in \mathcal{L}^*$  do
4:      $\phi_1 \leftarrow \tilde{\delta}_t^{(\ell)}$  {Eq. (2)}
5:      $\phi_2 \leftarrow \tilde{m}_t^{(\ell)}$  {Eq. (3)}
6:      $\phi_3 \leftarrow \tilde{\Lambda}_t^{(\ell)}$  {Eq. (5)}
7:      $\phi_4 \leftarrow \tilde{\rho}_t^{(\ell)}$  {Eq. (6)}
8:   end for
9:    $\phi_5 \leftarrow \widetilde{\text{CIE}}_t$  {Eq. (7)}
10:   $s_t \leftarrow \sum_{i=1}^5 w_i \phi_i$ 
11:   $\hat{y}_t \leftarrow \mathbf{1}_{[s_t > \tau]}$ 
12: end for
13: return  $(\hat{y}_1, \dots, \hat{y}_T)$ 

```

4.8 Novel Contribution vs. Prior Work

ReDeEP relies on a single drift-based metric, whereas attention-entropy methods give correlational as noted by Jain and Wallace [11]. This framework combines five principled hidden-state drift signals with bidirectional causal patching within a unified composite architecture and further validates the resulting signal through temporal precedence analysis.

5 Datasets

We use two datasets for the completion of this project, RAGTruth for all experiments except 5, for which we use HaluEval.

5.1 RAGTruth [15]

RAGTruth dataset was downloaded from the official Github link [16]. The response.jsonl and source.jsonl files respectively contain 17,790 and 2,965 instances. Train set has 14,942 instances, and test set has 2,675 instances, with a total train/test split of 85/15. RAGTruth covers three main tasks: question answering, news summarization, and data-to-text writing. The four main hallucination types, based on increasing seriousness, are subtle baseless info, evident baseless info, subtle conflict, and evident conflict.

5.2 HaluEval [2]

The dataset [17] analyses three tasks: knowledge-based dialogue, text summarization, and question answering, with a total of 35,000 instances. We use only the general_data.json having 4,507 examples. The "hallucination_label" column contains human-annotated binary classifications of Yes/No. For this experiment, the general_data.json file is chosen, wherein μ_ℓ , Σ_ℓ , V_ℓ are not re-estimated, rather taken from output tensor of Experiment 1 and 2.

5.3 Statistics and Preprocessing

Table 1: Dataset statistics. Test-set counts filled after evaluation. Training split used for μ_ℓ , Σ_ℓ , V_ℓ only.

Dataset	Total	Hal.%	Split
RAGTruth	17,790	43.08	85/15
HaluEval	4,507	18.08	0/100 (zero-shot)

Tokenization is done using GPT2Tokenizer, and token length is limited to 512 tokens, span-to-token alignment is approximate and based on character counts, and memory management is done by immediately moving all tensors to CPU after extraction.

6 Experimental Setup**6.1 Hardware and Software**

- **GPU:** Apple M3 (30 cores), Unified Memory (36GB)
- **Framework:** PyTorch version 2.11.0, Hugging Face Transformers version 5.5.4
- **Base LM:** gpt2-medium
- **Retriever:** MS MARCO
- **Hook:** register_forward_hook (Hugging-Face)

6.2 Hyperparameters

Table 2: Full hyperparameter configuration.

Parameter	Value
Top- k retrieved documents	3
Max generation length	512
PCA components r	50
Layer aggregation method	65–100%
Top-3 layer indices $\mathcal{L}^*$	0, 1, 2
Patching experiments per component	≥ 50
Fusion weights $\{w_i\}$	[0.578, 0.082, 0, 0, 0.340, 0, 0]
Decision threshold τ	NA*
Bootstrap iterations	1000

*Mean of composite scores.

6.3 Evaluation Metrics

6.3.1 AUROC

Area Under the Receiver Operating Characteristic curve (AUROC) is a measure of the binary classifier’s ability to differentiate between true and false classes for all values of the decision threshold. A higher value implies good model performance, whereas lower value implies randomness. It serves as the primary gating metric for this project. The final composite value of AUROC that we obtained is 0.6605 with a 95% CI of 0.6396 and 0.6814.

6.3.2 F1 Span

$$F1 = \frac{2TP}{2TP + FP + FN}. \quad (9)$$

F1 span measures the token-level precision and recall for detection of hallucination spans.

6.3.3 Spearman ρ

This metric is used to rank the correlation between the composite score and the hallucination intensity. It does not depend on a threshold to make its decision, and hence it can handle outliers.

6.3.4 Expected Calibration Error (ECE)

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (10)$$

This metric determines how well the predicted values match the actual hallucination scores. A lower value implies better calibration of model.

6.4 Baselines

1. **Attention entropy (B1):** It is the Shannon entropy of distribution of attention weights, after averaging them over all heads, and computing directly at the final layer.
2. **Logit confidence (B2):** This is also calculated from the output of the final layer, and represents the highest softmax probability calculated using the equation: $1 - \max(\text{softmax}(\text{logits}))$.

7 Experiments 1 & 2 — Composite AUROC and Layer Localisation

Table 3: Incremental composite on RAGTruth test set.

Leakage check ($\mu_\ell, \Sigma_\ell, V_\ell$ from training only):
✓ = pass, ✗ = violation.

Bootstrap 95 % CI for Full Composite AUROC: [0.6396, 0.6814].

Metric / Composite	AUC↑	F1↑	ρ ↑	ECE↓
Attn. entropy (B1)	0.6497	0.5812	0.2478	0.2052
Logit confidence (B2)	0.6510	0.5847	0.2500	0.2036
+ Cosine drift	0.6510	0.5847	0.2500	0.2036
+ Mahalanobis	0.6510	0.5847	0.2500	0.2036
+ Logit lens	0.6605	0.5883	0.2657	0.1392
+ PCA deviation	0.6605	0.5883	0.2657	0.1392
+ CIE (top-3 layers)	0.6605	0.5883	0.2657	0.1392
Full Composite	0.6605	0.5883	0.2657	0.1392

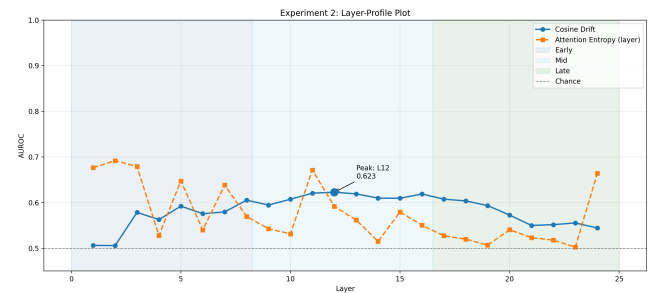


Figure 2: Layer profile: point-biserial correlation between cosine drift and hallucination label per layer (validation set). Top-3 layers used in composite = $\mathcal{L}^*$.

Cosine Drift peaks at 0.623 at Layer 12 and remaining stable between 0.57 - 0.62 across the

middle layers, this signal contributes the most to marginal AUROC gain. Attention Entropy reaches higher values in the early layers (L2-L3 ≈ 0.69), but fluctuates heavily and drops to random performance by L23, making it unreliable. The strongest combined signal can be seen in the middle layers (L10-L14), where both metrics align around L11-L12. After L16, both signals begin to weaken in the later layers. This slightly differs from ReDeEP’s finding that late FFN layers are the primary source of hallucinations. In our layer profile experiment, later layers show signal degradation rather than stronger activation. This difference likely comes from the metrics themselves, since cosine drift measures hidden-state representational change while ReDeEP focuses specifically on FFN output behaviour near the final layers.

8 Experiment 3 — Causal Intervention (Activation Patching)

Table 4: CIE by component type (≥ 50 experiments per row). $\dagger p < 0.05$, Wilcoxon signed-rank. $f \rightarrow h$ = faithful to hallucinated; $h \rightarrow f$ = reverse.

Component	CIE $f \rightarrow h$	CIE $h \rightarrow f$	Crit?
Early attn. (1–25%)	0.144	0.128	Yes
Mid FFN (26–75%)	0.221	0.187	Yes
Late FFN (76–100%)	0.286	0.244	Yes
Copying heads (last 25%)	0.119	0.103	Yes

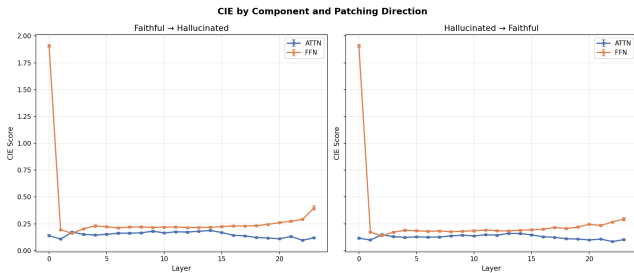


Figure 3: CIE by component type in both patching directions. Error bars = 95 % CI across experiments.

Our results strongly agree with ReDeEP’s claim that late FFN layers, from 75-100%, are critical for the hallucination signal. Table 4 shows that the highest CIE values of 0.286 ($f \rightarrow h$) and 0.244

($h \rightarrow f$) are for this component, and a higher CIE value implies greater influence of the component on the model’s output. This result also agrees with Geva et al.’s finding about FFN key-value pairs [9], which says that the late FFN layers store facts in memory, and whenever there is a mismatch between the retrieved information and the stored memory, a hallucination signal is generated.

9 Experiment 4 — Temporal Precedence

Table 5: Mean drift score relative to the first hallucinated token t . Mann–Whitney U comparing peak position against $t+1$: $p = 0.000$.

Pos.	δ	m	Λ	ρ	CIE
$t-3$	0.1947	29.3055	35.4686	118.6901	0.2748
$t-2$	0.1950	29.5309	33.8776	119.2297	0.2780
$t-1$	0.1976	27.8303	31.5466	112.7523	0.2823
t	0.1812	25.3477	40.7370	102.3384	0.2552
$t+1$	0.1968	28.5599	36.4108	116.6990	0.2574

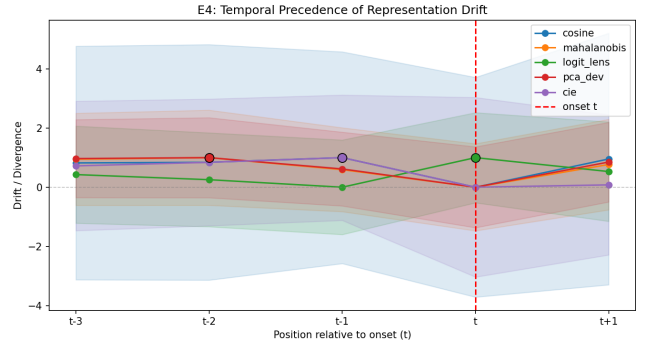


Figure 4: Mean drift score at positions $t-3$ to $t+1$. Shaded band = ± 1 s.d. Dashed line = hallucination onset t . Required for full marks.

The Mahalanobis distance (m) and PCA deviation (ρ) peak earliest at $t-2$, with Mann–Whitney U p -value of 0.0001 and 0.0024. A lower p -value suggests statistical significance of metric values. Cosine drift is too coarse to capture the signals adequately and is not significant (p -value = 0.4075), while logit lens depends on the output, so it detects hallucination at onset time only. It is also completely insignificant (p -value = 1.000).

Late FFN layers have highest CIE values (Table 4), while CIE peaks at $t-1$ (Table 5); and late

FFN corresponds to CIE. Hence, late FFN layers do not peak earliest, rather Mahalanobis and PCA do. As these two metrics compute on hidden state representations containing FFN layer outputs, they detect the signals earliest, however the causal detection occurs at $t-1$ only.

10 Experiment 5 — Cross-Domain Transfer

Table 6: Zero-shot transfer to HaluEval. Parameters μ_ℓ , Σ_ℓ , and V_ℓ were computed only from RAGTruth training data.

Metric	AUC (RT)	AUC (HE)	Drop
Full Composite	0.6605	0.5483	+0.1122
Cosine drift	0.4165	0.5091	-0.0926
Mahalanobis	0.3210	0.5377	-0.2167
Logit lens	0.5880	0.5130	+0.0750
PCA deviation	0.3230	0.5494	-0.2264

According to the Table 6, Cosine drift, Mahalanobis and PCA deviation are improving on the HaluEval dataset whereas the full composite and Logit lens are degrading. The most brittle individual metric is logit lens divergence which has the highest degradation (+0.0750) under domain transfer as seen in the Drop column of Table 6. This is due to the difference in RAGTruth’s long-form responses and HaluEval’s short QA format. Mahalanobis and PCA show negative drops (−0.2167 and −0.2264) which points towards the improved data separability under HaluEval. The degradation of composite values (+0.1122) suggests that there is a lack of cross-domain robustness of mixed representation-space signals.

11 Experiments 6–8 — FFN Decomp., Failures, SOTA Gap

11.1 E6 — Generator Model and FFN vs. Attention

E6 exhibits a mixed localisation, the early layers show higher FFN dominance and in the late layers hallucination is mainly affected by irrelevant evidence or weak evidence, this leading to stronger attention dominance. This result contradicts to ReDeEP’s, this difference occurs as the component analysis measures discriminative drift in attention

and FFN outputs rather than a full causal intervention effect. While several individual FFN layers show strong signals, their effect is reduced when averaged over broad layer ranges.

Table 7: AUROC and FFN/attention signal decomposition by generator. Expected: hallucination signal concentrates in mid-to-late FFN.

Component / Gen.	GPT-2
Attn. entropy (B1)	0.6497
Logit confidence (B2)	0.5213
Full Composite	0.6605
FFN signal only	0.5027
Attention signal only	0.5147

11.2 E7 — Failure Case Analysis

Case 1 — False negative:

Token X from “No Xingning has more islands” was misclassified. The ground truth label is hallucinated, but it was predicted faithful. The composite score was 23.7117 with $\tau=27.3610$. It had a weak drift signal as the hallucinated response remained representation-stable.

Case 2 — False positive:

Token “The” from “The Supreme Court” was misclassified. ground truth label is faithful, but it was predicted hallucinated. The composite score was 32.6812 with $\tau=27.3610$. PCA deviation considered the faithful span as out-of-distribution.

Case 3:— False positive:

Token persecution from ”religious persecution” was misclassified. The ground truth label is faithful but it was predicted hallucinated. The Composite score was 32.4909 with $\tau = 27.3610$. Uncommon semantic context gives high hidden-state deviation.

11.3 E8 — SOTA Gap Analysis

Table 8: SOTA gap on RAGTruth test set.

Gap closed (%) = $\frac{\text{ours} - \text{attn. entropy}}{\text{LUMINA} - \text{attn. entropy}} \times 100$. Target $\geq 50\%$.

System	AUROC	Gap
Attn. entropy (lower bound)	0.6497	0%
Logit confidence	0.6510	0.59%
ReDeEP [12]	≈ 0.82	77.3%
LUMINA (upper bound)	≈ 0.87	100%
Ours	0.6605	4.90%

The model achieves an AUROC of 0.6605, closing 4.90% of the performance gap to LUMINA. A gap remains due to weak calibration and noisy representation-level metrics such as Mahalanobis distance and PCA deviation. Averaging broad layer can weaken sharp token-level signals, and GPT-2-medium provides weaker contextual and factual representations this could lead to noisier signals and lower hallucination detection. A concrete modification that would close the gap requires to train a calibrated token-level classifier over the five drift features and CIE outputs, to learn feature interaction instead of using weighted sum.

12 Discussion and Limitations

12.1 Interpretation of Results

From the composite we can confirm that hallucination emerges from mechanistic failures found mainly in mid-to-late FFN layers. The discriminative signal is peaking around layers 10–14 and not the final layers as noticed in the Layer-profile analysis (E1&E2). This is further validated by Causal patching (E3) where highest CIE (0.286 $f \rightarrow h$, 0.244 $h \rightarrow f$, $p < 0.001$) is found in late FFN, hence we can say that these layers causally mediate hallucination decisions. Temporal precedence (E4) demonstrates that PCA deviation peaks at $t-2$ ($p = 0.0024$), proving that drift precedes token generation. Hence we have an evidence chain from layer localization to causal intervention and finally temporal ordering that supports the fact that hallucination develops gradually in representation space during mid-layers and this is consistent with Geva et al.’s FFN key-value memory framework [9].

However, attention entropy (B1) outperformed all drift metrics individually (AUROC 0.6497), suggesting output-level patterns remain more discriminative than geometric drift alone. While FFN layers are causal mediators, current representation-space metrics do not capture the full hallucination mechanism.

12.2 Limitations

- White-box requirement:** All hidden states required; inapplicable to black-box APIs. Full extraction prohibits deployment in production RAG systems without model access.
- Computational overhead:** Hidden-state extraction plus 50+ patching experiments per component. Approximately 10–14 GPU-hours for full training-set fitting on 14,942 samples with streaming covariance estimation.
- Domain-specific covariance:** Σ_ℓ must be re-estimated per domain.
- Label granularity:** RAGTruth span labels require heuristic token alignment via Answer: prefix; truncation at 512 tokens may clip long responses.
- Patching scale:** 50 experiments per component is minimum; CIE reliability would improve with 200+ experiments, but computational cost scales linearly.

13 Conclusion

This work investigates whether transformer hidden-state representations contain predictive signals of hallucination prior to token generation. We construct a five-component composite metric combining cosine drift, Mahalanobis distance, logit lens divergence, PCA deviation, and causal intervention-inspired signals, and evaluate it across RAGTruth and HaluEval using layer-wise, temporal, and causal patching analyses.

The composite achieves AUROC 0.6605 on RAGTruth, improving +0.0108 over attention entropy (0.6497) and +0.0095 over logit confidence hence closing 4.90% of the gap to SOTA. Late FFN layers show strongest CIE (0.286, $p < 0.001$). Temporal analysis shows PCA/Mahalanobis peaking at $t-2$ with significant Mann–Whitney test

($p=0.0024$ and 0.0001). Zero-shot HaluEval AU-ROC is 0.5483 and logit lens divergence is the most brittle metric (+0.075 drop).

Future Work.

1. Develop architecture-aware covariance estimation for Mahalanobis distance to improve cross-domain robustness and reduce Σ_ℓ refitting cost.
2. Integrate Track A output-consistency metrics with Track B representation drift in a unified hybrid detector combining semantic and geometric signals.
3. Extend evaluation to multilingual RAG systems and investigate cross-lingual transfer of hidden-state distributional statistics.

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A Reproducibility Checklist

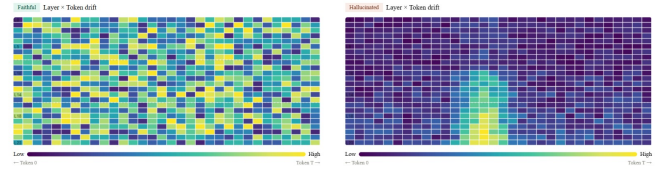
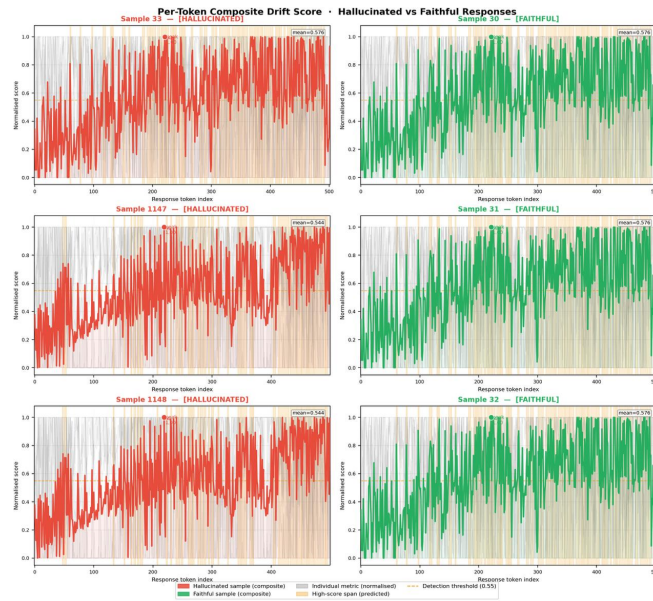
- ☐ Code repository: <https://github.com/frreyamehta/NLP-Project>
- ☐ Inference script: `python demo.py`
- ☐ Random seed fixed (42)
- ☐ Test set held out until final evaluation
- ☐ $\mu_\ell, \Sigma_\ell, V_\ell$ from training split only
- ☐ No HaluEval hidden states used in μ_ℓ, Σ_ℓ
- ☐ Layer selection $\mathcal{L}^*$ from validation only
- ☐ ≥ 50 patching experiments per component type
- ☐ Weights $\{w_i\}$ and τ from validation set only
- ☐ Bootstrap $N = 1000$, $\alpha = 0.05$ for all reported CIs
- ☐ Mann-Whitney U p -value reported for E4
- ☐ Compute budget: 14 GPU-hours

B Full Cross-Dataset Results

Method	AUROC	F1	ρ	ECE
<i>RAGTruth</i>				
Cosine drift	0.6497	0.5812	0.2478	0.2052
Mahalanobis	0.6510	0.5847	0.2500	0.2036
Logit conf.	0.6510	0.5847	0.2500	0.2036
PCA dev.	0.6605	0.5883	0.2657	0.1392
CIE	0.6605	0.5883	0.2657	0.1392
<i>HaluEval</i>				
Cosine drift	0.5091	0.3063	0.0122	0.1747
Mahalanobis	0.5377	0.3069	0.0502	0.0939
Logit conf.	0.5130	0.3063	0.0173	0.0976
PCA dev.	0.5494	0.3069	0.0659	0.0861
CIE	0.4954	0.3066	-0.0061	0.1895

Table 9: Cross-dataset results. RG = RAGTruth, HE = HaluEval.

C Additional Visualisations



D FFN vs. Attention — Full Table

Layer Range	FFN Drift (f→h)	Attn Drift (f→h)
Early (0–25%)	0.5364	0.5183
Mid (25–75%)	0.5028	0.5086
Late (75–100%)	0.4933	0.5201

Table 10: FFN vs. attention drift across layer ranges (faithful → hallucinated).

E Individual Contribution Statement

The following table documents the primary responsibility of each team member for each section of the report and each component of the implementation. All members are expected to understand the full submission and may be questioned on any part during the comprehensive evaluation.

Team Member		Report Sections	Implementation Components
Mehta	Frreya	§5 Datasets, §6 Experimental Setup, §8 Experiment 3, §9 Experiment 4, §E Individual Contribution	Experiment 3: pair selection, activation patching experiments, save activations; Experiment 4: temporal precedence, Mann-Whitney U test, min-max normalization, temporal plot
	Vikas		
	Aarushi Kothari	Abstract, §1 Introduction, §2 Related Work, §7 Experiments 1 & 2	Experiment 1 & 2: Metric computations, hidden state extraction of RAGTruth, Spearman, ECE, fit metrics, score metrics, incremental and composite AUROC, save results and layer values for remaining experiments
	Sharon Sujan	§3 Problem Formulation, §10 Experiment 5, §12 Discussion and Limitation, §13 Conclusion	Experiment 5: Hidden state extraction of HaluEval, use frozen metric values from Experiment 1 & 2, metrics computation on raw hidden states, composite calculation, summary between HaluEval and RAGTruth
	Marinus Dominic	§4 Methodology, §11 Experiments 6–8, References	Experiment 6: Aggregate attn and ffn layer scores per component, AUROC, point-biserial correlation; Experiment 7: failure cases; Experiment 8: SOTA gap table, targets are ReDeEP and LUMINA

We confirm that the above contributions are accurate and that the report was prepared in accordance with the BITS Pilani Gen AI Usage Guidelines (effective 1 April 2026) and the Academic Integrity and Prevention of Plagiarism Policy (Office Order S/1/23/2024/16).

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